Keshav Sharma

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EDUCATION

Vellore Institute of Technology — **Chennai, Tamil Nadu**

Bachelor of Technology in Computer Science (Data Science Specialisation) — *Graduation: May 2027*

SKILL SUMMARY

Programming Languages: Python, Java, C++, SQL
Core Computer Science: Data Structures & Algorithms, Object-Oriented Programming (OOP), Operating Systems, Computer Networks, DBMS
Data Science & Machine Learning: NumPy, Pandas, Scikit-learn, Data Preprocessing, Exploratory Data Analysis, Anomaly Detection, Model Evaluation
Backend & Software Development: FastAPI, Flask, REST APIs, Git, GitHub
Databases: MySQL, SQLite, MongoDB, Redis
Tools & Visualization: Power BI, Matplotlib, Plotly, Jupyter Notebook

EXPERIENCE

AI for Sustainability Virtual Intern* |LINK** — ***July – Sep, 2026

- Completed an AI-focused virtual internship covering Generative AI, Prompt Engineering, RAG, Agentic AI, and Responsible AI, with hands-on exposure to enterprise AI workflows and LLM applications.
- Applied RAG and semantic search concepts using document ingestion, embeddings, vector databases, and retrieval pipelines to improve the grounding and reliability of LLM responses.
- Built an Energy Anomaly Detector using Python, Pandas, NumPy, and Isolation Forest, detecting 9 anomalies across 6,918 test hours (0.13%), with automated email alerts, privacy safeguards, transparent scoring, and rate-limited notifications.

PROJECTS

***Energy Anomaly Detector*| LINK**

- Developed a Python pipeline to process electricity-consumption time series using hourly resampling, temporal feature engineering, rolling baselines, and Isolation Forest.
- Automated anomaly logging, visual reporting, and email alerts, with rate limiting to control alert frequency.
- Evaluated detections against a 95th-percentile threshold, identifying 9 anomalies across 6,918 test hours.

***Real-Time Stock Market Anomaly Detector* |LINK**

- Engineered a real-time market analytics pipeline processing 2M+ NSE records, combining price signals and financial-news sentiment to detect unusual market behaviour.
- Developed and evaluated Isolation Forest and LSTM Autoencoder models, achieving 88% precision and 85% recall on validation data for anomaly detection.
- Optimised the analytics pipeline to reduce detection latency by 35%, enabling near real-time monitoring and faster identification of unusual market events.

***RedisLite* |LINK**

- Built a Redis-inspired in-memory data store in Python with a custom RESP parser, multithreaded TCP server, thread-safe storage, and O(1) LRU caching.
- Implemented TTL expiration, write-ahead logging, and crash recovery across a four-layer architecture.
- Added 46 automated PyTest unit tests and benchmarked 619K SET operations/sec in-memory and 2.8K operations/sec with durable WAL persistence.

***Docwise* |LINK**

- Built an AI-powered PDF question-answering system using Google Gemini and RAG, enabling contextual question answering over documents exceeding 200 pages.
- Developed a document processing and retrieval pipeline combining PDF extraction, chunking, embeddings, and semantic retrieval, achieving 95%+ query accuracy.
- Integrated FastAPI and LLM inference to deliver an end-to-end AI application for document understanding and conversational querying.

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional |**LINK** 2026
- Deloitte Australia Data Analytics Job Simulation |**LINK** 2025
- AWS APAC Solutions Architecture virtual experience program | **LINK** 2025